Overall Survival Prediction in Adult Myeloid Leukemia: Kaplan–Meier, Cox Regression and Machine-Learning Survival Models

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Abstract

This report documents my work on the QRT × Institut Gustave Roussy data challenge on Challenge Data (ENS Paris), whose goal is to rank adult patients with myeloid leukemia by their risk of death, using clinical measurements, cytogenetic reports and somatic gene mutations. Models are scored with the inverse probability of censoring weighted (IPCW) concordance index truncated at seven years. The data audit shows that the 1,193 test patients all come from a single clinical centre that is absent from the 23 training centres, and that this centre's patients are clinically sicker, so model validation will hold out whole centres. Kaplan–Meier analysis gives a median overall survival of 3.15 years, with strong separation by bone-marrow blasts, *TP53* and *SF3B1* mutations and monosomy 7 (log-rank $p < 0.001$ for all). The Kaplan–Meier estimator, a log-rank test and both concordance indices were computed by hand and match standard software. Single variables reach an IPCW C-index of at most 0.616, which is the baseline that multivariable models must beat.

1 Introduction

Myeloid leukemias are blood cancers in which abnormal immature cells (blasts) accumulate in the bone marrow. Estimating a patient's prognosis guides treatment: low-risk patients may receive supportive care, while high-risk patients may be prioritised for intensive treatment such as stem cell transplantation.

The outcome is *overall survival* (OS): the time from diagnosis to death or to the last follow-up. When a patient is still alive at the last follow-up, the true survival time is unknown; we only know it is longer than the observed time. Such observations are called *right-censored*, and handling them correctly is the purpose of survival analysis.

The project follows a staged plan: Kaplan–Meier estimation and log-rank tests, Cox proportional hazards regression, and machine-learning survival models (gradient-boosted survival and random survival forests), all compared under the challenge metric with honest cross-validation. A similar crowdsourced effort, the DREAM 9 AML Outcome Prediction Challenge, also scored survival models with a concordance index [1].

2 Data

The data come from 24 clinical centres and are split into the five files described in Table 1. The `ID` column links the files.

Table 1: The five challenge files.

File	Rows	Content
`clinical_train.csv`	3,323	one row per patient: centre, blood counts, karyotype
`clinical_test.csv`	1,193	same, for test patients
`molecular_train.csv`	10,935	one row per somatic mutation
`molecular_test.csv`	3,089	same, for test patients
`target_train.csv`	3,323	`OS_YEARS`, `OS_STATUS`

Clinical variables. `BM_BLAST` (bone-marrow blasts, %), `WBC` (white blood cells, G/L), `ANC` (absolute neutrophil count, G/L), `MONOCYTES` (G/L), `HB` (haemoglobin, g/dL), `PLT` (platelets, G/L) and `CYTOGENETICS`, a free-text karyotype written in ISCN notation (for example `46,xy` for a normal male karyotype).

Molecular variables. Each row is one somatic mutation, with its chromosomal position, reference and alternate bases, the gene, the protein change, the broad effect, the variant allele fraction (`VAF`, the share of sequencing reads carrying the mutation) and, not mentioned in the challenge description, the sequencing read depth (`DEPTH`).

3 The evaluation metric

For each patient i we observe a time T_i (`OS_YEARS`), an event indicator Δ_i (1 = death, 0 = censored), and the model gives a risk score η_i.

Harrell's C-index. A pair of patients (i, j) is *comparable* when we know who died first, that is, when $T_i < T_j$ and patient i died ($\Delta_i = 1$). The pair is *concordant* when the model agrees, $\eta_i > \eta_j$. Harrell's C-index [2] is

$$C = \frac{\text{number of concordant comparable pairs}}{\text{number of comparable pairs}}. \tag{1}$$

A value of 0.5 corresponds to random guessing and 1 to a perfect ranking. Pairs with equal risk scores count as one half.

IPCW C-index (Uno's C). Patients who die late are less likely to be observed dying, because many patients are censored before then. Harrell's C therefore over-represents early deaths. Uno *et al.* [3] correct this by weighting each comparable pair by $1/\hat{G}(T_i)^2$, where $\hat{G}(t)$ is the Kaplan–Meier estimate of the probability of *not yet being censored* at time t. With the truncation horizon $\tau = 7$ years used by the challenge,

$$C_\tau = \frac{\displaystyle\sum_i \sum_j \Delta_i \, \hat{G}(T_i)^{-2} \, \mathbf{1}(T_i < T_j,\ T_i < \tau)\, \mathbf{1}(\eta_i > \eta_j)}{\displaystyle\sum_i \sum_j \Delta_i \, \hat{G}(T_i)^{-2} \, \mathbf{1}(T_i < T_j,\ T_i < \tau)}. \tag{2}$$

In plain language: a death at a time when few patients remain uncensored is rare to observe, so it counts more. Only the ordering of the scores η_i matters. The challenge computes C_τ with `scikit-survival` [4].

4 Data audit (Stage 1)

4.1 Linking and outcome

All IDs are unique, every mutation row belongs to a known patient, and no patient appears in both training and test sets. However, 150 training patients have no recorded outcome and are excluded from modelling. Table 2 summarises the remaining outcomes and Figure 1 shows the observed times.

Table 2: Outcome of the training patients.

Quantity	Value
Training patients	3,323
Without outcome (excluded)	150
With outcome	3,173
Deaths ($\Delta = 1$)	1,600 (50.4%)
Censored ($\Delta = 0$)	1,573 (49.6%)
Median observed time (years)	1.65
Maximum observed time (years)	22.04
Observed time exactly 0 (all censored)	53
Observed beyond the 7-year horizon	172

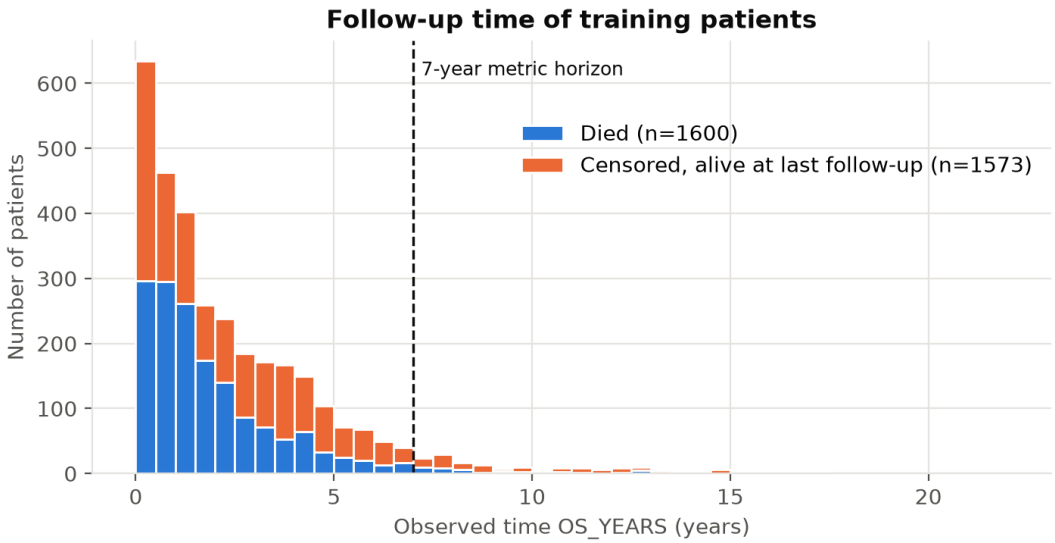


Figure 1: Observed time of the 3,173 training patients with an outcome, split into deaths and censored patients. The dashed line marks the 7-year horizon of the metric.

About half the patients are censored, so simply discarding them or treating their observed time as a death time would badly bias any model. Most deaths occur within the first two years, and only 172 patients are followed beyond seven years, which justifies the truncation of the metric.

4.2 Clinical centres: the test set is a new hospital

The 3,323 training patients come from 23 centres, while all 1,193 test patients come from one centre, `KYW`, that does not appear in training (Figure 2). Death rates also vary widely between

training centres, from 11% (ROM) to 92% (MUV), reflecting different patient populations and follow-up lengths.

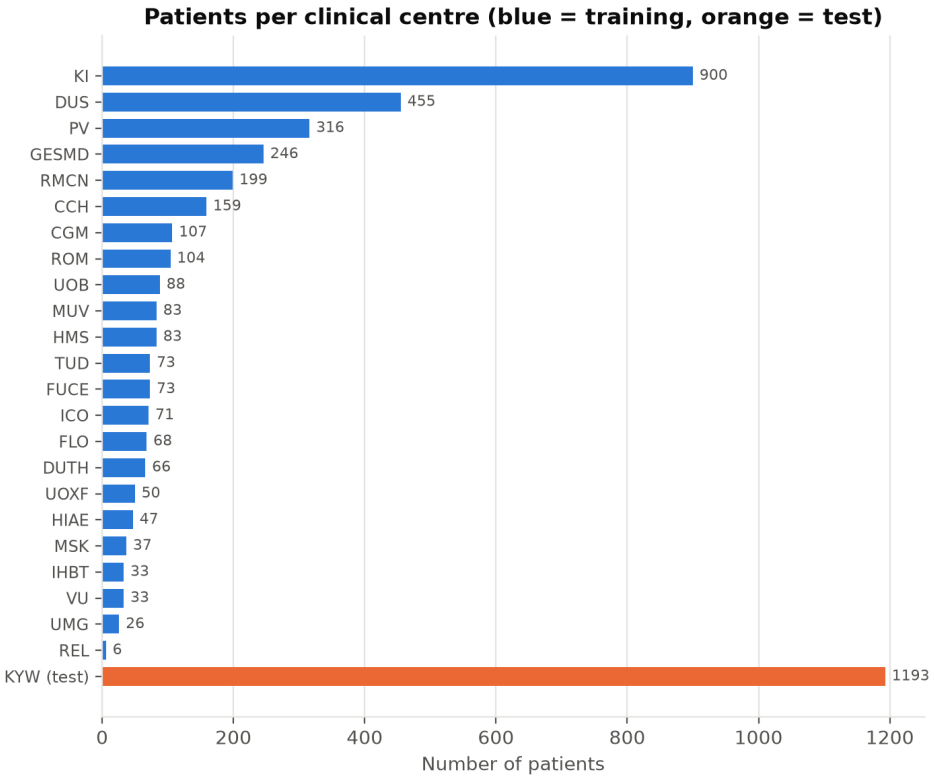


Figure 2: Number of patients per clinical centre. The test centre KYW (orange) contributes no training patients.

This has a direct consequence for validation. A random K-fold split mixes patients from the same centre into training and validation folds, so the model is always validated on centres it has already seen. The real task is to predict for an *unseen* centre. Cross-validation will therefore be grouped by centre, and the CENTER variable cannot be used as a predictor.

4.3 Clinical variables: training versus test

Table 3 and Figures 3 and 4 compare the clinical variables in the two sets.

Table 3: Missing values and medians of the clinical variables.

	Missing (%)		Median	
Variable	Train	Test	Train	Test
BM_BLAST (%)	3.3	9.6	3.00	6.00
WBC (G/L)	8.2	9.4	4.10	2.90
ANC (G/L)	5.8	11.8	2.00	1.30
MONOCYTES (G/L)	18.1	74.0	0.37	0.20
HB (g/dL)	3.3	9.3	9.70	8.40
PLT (G/L)	3.7	9.6	123.00	74.00
CYTOGENETICS	11.6	9.7	–	–

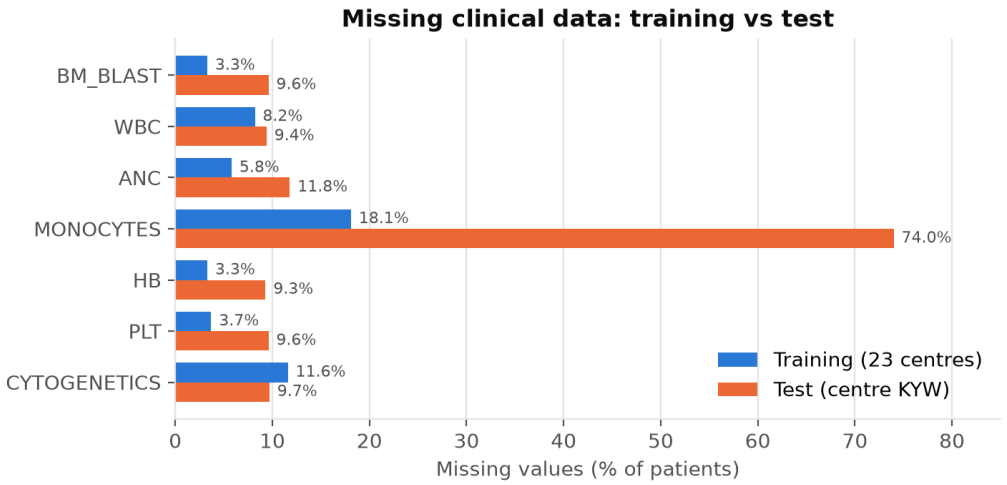


Figure 3: Percentage of missing values in each clinical variable.

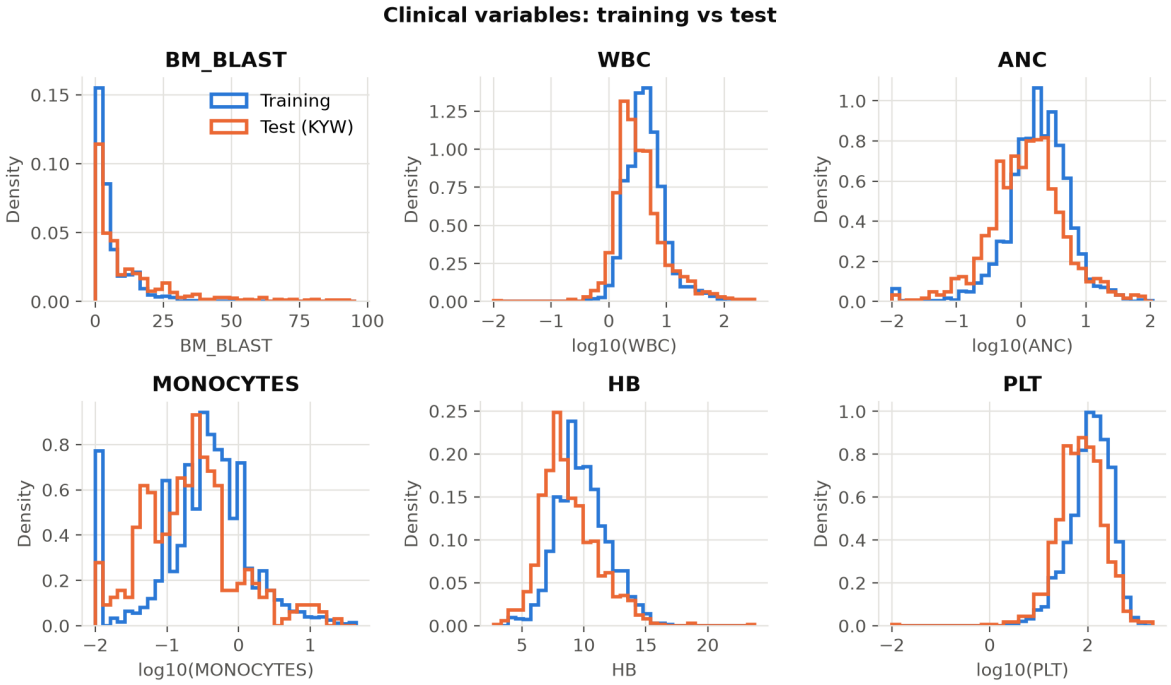


Figure 4: Distributions of the clinical variables in training (blue) and test (orange). Each histogram is scaled to area one so that sets of different sizes can be compared. Skewed blood counts are shown on a $\log_{10}$ scale; the spike at -2 for monocytes corresponds to recorded values of zero.

The test patients are clinically sicker: lower haemoglobin and platelets, lower neutrophils, and a much larger share with bone-marrow blasts above 20% (17.6% versus 4.2%). Monocytes are missing for 74% of test patients (883 of 1,193), so a model that leans on that variable would be fragile on the test set: all 883 would receive the same imputed value, and the training median (0.37) is above the test centre's own median (0.20).

4.4 Gene mutations

Training patients carry on average 3.29 recorded mutations and test patients 2.59; 297 training and 139 test patients have none, which is genuine information rather than missing data. Training data contain 124 distinct genes and test data 83, of which 19 never appear in training. Of the training rows, 114 lack a chromosomal position: they are partial tandem duplications of *MLL* (89) and internal tandem duplications of *FLT3* (25), which are structural events rather than single-base changes. In 1,274 cases the same gene is mutated more than once in one patient.

Table 4: Ten most frequently mutated genes in training: percentage of patients with at least one mutation.

Gene	Train (%)	Test (%)
TET2	31.1	17.5
ASXL1	26.9	12.7
SF3B1	22.2	7.7
SRSF2	17.2	9.1
DNMT3A	16.1	11.1
RUNX1	13.9	13.7
TP53	11.4	25.7
STAG2	9.1	6.7
U2AF1	8.6	8.4
EZH2	6.5	6.7

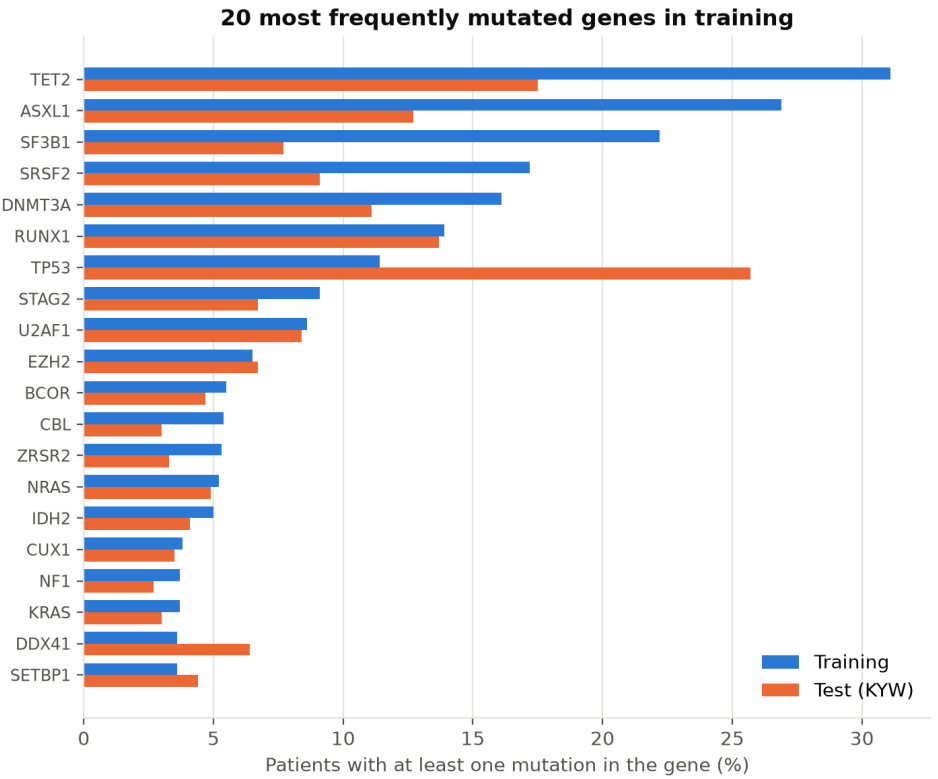


Figure 5: The twenty most frequently mutated genes in training, with their frequency in the test centre.

The most striking difference is *TP53*, mutated in 25.7% of test patients against 11.4% in

training (Table 4, Figure 5). *TP53* mutations are associated with poor prognosis in myeloid neoplasms, which is consistent with the sicker clinical profile of the test centre. By contrast, *SF3B1* mutations, typically associated with lower-risk disease, are almost three times rarer in the test set.

4.5 Cytogenetics text

In training, karyotypes are written in lower case (`46,xy`); in the test set they are written in upper case (`46,XY`), and 241 test patients have the single word `Normal`. Any parsing must first convert text to lower case and map `Normal` to a normal karyotype; otherwise every test karyotype would look unfamiliar to the model.

4.6 Consequences for modelling

1. Exclude the 150 training patients without outcome.
2. Validate with cross-validation grouped by centre, and do not use `CENTER` as a predictor.
3. Prefer features that are measured consistently in both sets; treat `MONOCYTES` with caution.
4. Normalise the cytogenetics text before extracting features.
5. Encode “no mutation found” as zero, not as missing.

5 Survival analysis without a model (Stage 2)

5.1 The Kaplan–Meier estimator

Let $t_1 < t_2 < \dots$ be the distinct times at which at least one death occurs. At each t_k let n_k be the number of patients still *at risk* just before t_k (neither dead nor censored yet) and d_k the number of deaths at t_k. The Kaplan–Meier estimator [5] of the survival function is

$$\hat{S}(t) = \prod_{k:\ t_k \le t} \left(1 - \frac{d_k}{n_k}\right). \tag{3}$$

In plain language: to survive past t_k you must survive every earlier death time, and at each one the chance of surviving, given you were still alive, is the share of the at-risk patients who did not die. Censored patients never make the curve drop; they simply leave the risk set, so they count fully for as long as they were observed.

Table 5 shows the first six steps. The first death occurs on day one ($t = 0.00274$ years) with $n = 3{,}120$ patients at risk: this is $3{,}173 - 53$, because the 53 patients with an observed time of zero were all censored at that time and left the risk set immediately. The first two steps are

$$\hat{S}(0.00274) = 1 - \tfrac{1}{3120} = 0.99968,$$
$$\hat{S}(0.00822) = 0.99968 \times \left(1 - \tfrac{2}{3114}\right) = 0.99968 \times 0.99936 = 0.99904.$$

My hand implementation of Equation (3) agrees with the `lifelines` library [7] at every death time to within 2×10^{-14}.

Table 5: First six steps of the Kaplan–Meier estimator, computed by hand.

t_k (years)	n_k	d_k	$1 - d_k/n_k$	$\hat{S}(t_k)$
0.00274	3120	1	0.99968	0.99968
0.00822	3114	2	0.99936	0.99904
0.01644	3107	1	0.99968	0.99872
0.02192	3103	2	0.99936	0.99807
0.02466	3098	1	0.99968	0.99775
0.02740	3095	2	0.99935	0.99711

The median overall survival, the first time at which $\hat{S}(t) \leq 0.5$, is **3.15 years**. Table 6 and Figure 6 give the whole curve.

Table 6: Kaplan–Meier overall survival with 95% confidence intervals.

Years	$\hat{S}(t)$	95% CI
1	0.790	0.774–0.804
2	0.614	0.595–0.633
3	0.509	0.489–0.529
5	0.364	0.342–0.386
7	0.275	0.250–0.299

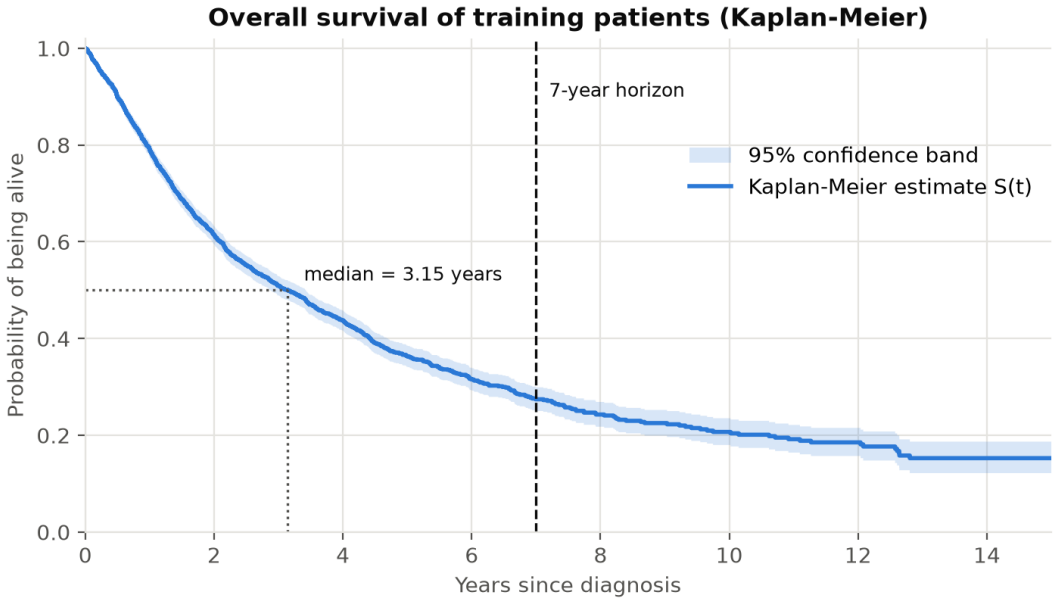


Figure 6: Kaplan–Meier estimate of overall survival for the 3,173 training patients, with a 95% confidence band.

5.2 Comparing groups: the log-rank test

To ask whether two groups have different survival, the log-rank test [6] compares, at each death time, the deaths observed in group 1 with the deaths it would receive if the groups were identical.

With n at risk in total, n_1 in group 1, $n_2 = n - n_1$, d deaths in total and d_1 in group 1,

$$E_{1k} = d\,\frac{n_1}{n}, \qquad V_k = \frac{n_1\, n_2\, d\,(n-d)}{n^2(n-1)}, \qquad \chi^2 = \frac{\left(\sum_k (d_{1k} - E_{1k})\right)^2}{\sum_k V_k}, \tag{4}$$

which follows a χ^2 distribution with one degree of freedom when the groups do not differ. In plain language: group 1 should get "its fair share" n_1/n of the deaths at each time; χ^2 measures how far the total gap is from zero, in units of its own variability.

Worked example: *TP53*. Summing over all death times, the 368 patients with a *TP53* mutation had $O_1 = 271$ observed deaths against $E_1 = 112.13$ expected, with total variance $V = 103.17$. Hence

$$\chi^2 = \frac{(271 - 112.13)^2}{103.17} = \frac{158.87^2}{103.17} = 244.64, \qquad p = 3.8 \times 10^{-55},$$

identical to the `lifelines` result. *TP53*-mutated patients died more than twice as often as expected under "no difference".

Table 7 and Figure 7 show four comparisons. Monosomy 7 was detected in the karyotype text as "`-7`" after a comma, slash or the start of the string and not followed by another digit; 79 patients without a blast value are left out of the blast comparison.

Table 7: Kaplan–Meier summaries and log-rank tests by group. All $p < 0.001$.

Comparison	Group	n	Deaths	Median (y)	$\hat{S}(3)$	χ^2 (df)
Blasts	$< 5\%$	2015	841	4.47	0.620	
	5–20%	942	622	1.68	0.326	272.1 (2)
	$> 20\%$	137	89	1.19	0.238	
TP53	not mutated	2805	1329	3.59	0.549	244.6 (1)
	mutated	368	271	1.00	0.207	
Monosomy 7	no	3008	1477	3.41	0.528	139.5 (1)
	yes	165	123	0.95	0.169	
SF3B1	not mutated	2446	1289	2.57	0.463	51.8 (1)
	mutated	727	311	4.56	0.665	

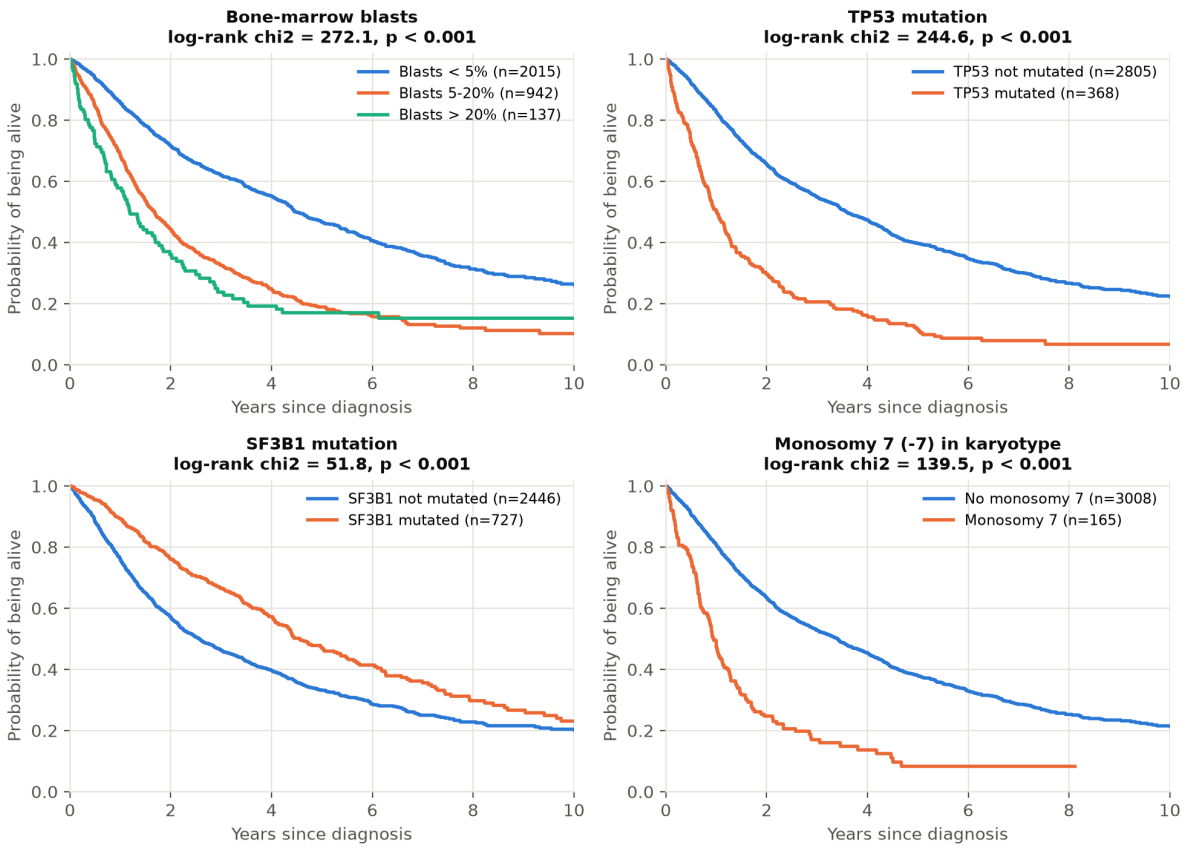


Figure 7: Kaplan–Meier curves by bone-marrow blast level, *TP53* mutation, *SF3B1* mutation and monosomy 7.

Higher blasts, *TP53* mutation and monosomy 7 all shorten survival sharply; the median survival of *TP53*-mutated patients is one year against 3.59 years for the rest. *SF3B1* is protective: mutated patients have a median survival of 4.56 years against 2.57. These directions agree with established clinical risk stratification and with the sicker profile of the test centre noted in Stage 1.

5.3 The censoring curve

Swapping the roles of death and censoring in Equation (3), so that the "event" is being censored, gives the *reverse Kaplan–Meier* estimate $\hat{G}(t)$: the probability that a patient is still being followed at time t. This is exactly the $\hat{G}$ in the metric (Equation (2)).

Table 8: Censoring curve and the resulting metric weight of a death at time t.

t (years)	$\hat{G}(t)$	weight $1/\hat{G}(t)^2$
1	0.827	$1/0.827^2 = 1.46$
3	0.613	$1/0.613^2 = 2.66$
5	0.347	$1/0.347^2 = 8.32$
7	0.198	$1/0.198^2 = 25.61$

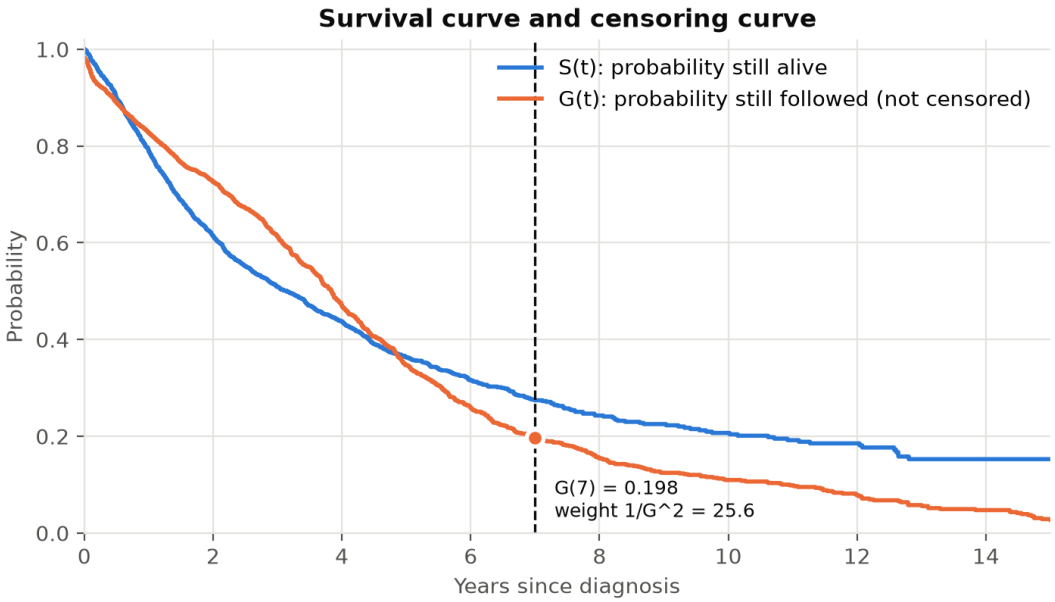


Figure 8: Survival curve $\hat{S}(t)$ and censoring curve $\hat{G}(t)$. By seven years only 19.8% of patients are still being followed.

Table 8 shows the practical effect: a pair started by a death just before seven years weighs about $25.6/1.46 \approx 17.5$ times as much as one started by a death at one year. A model that ranks late deaths poorly is therefore penalised far more by the challenge metric than by Harrell's C.

5.4 The metric by hand: a six-patient example

Figure 9 shows six patients with observed times, statuses and risk scores.

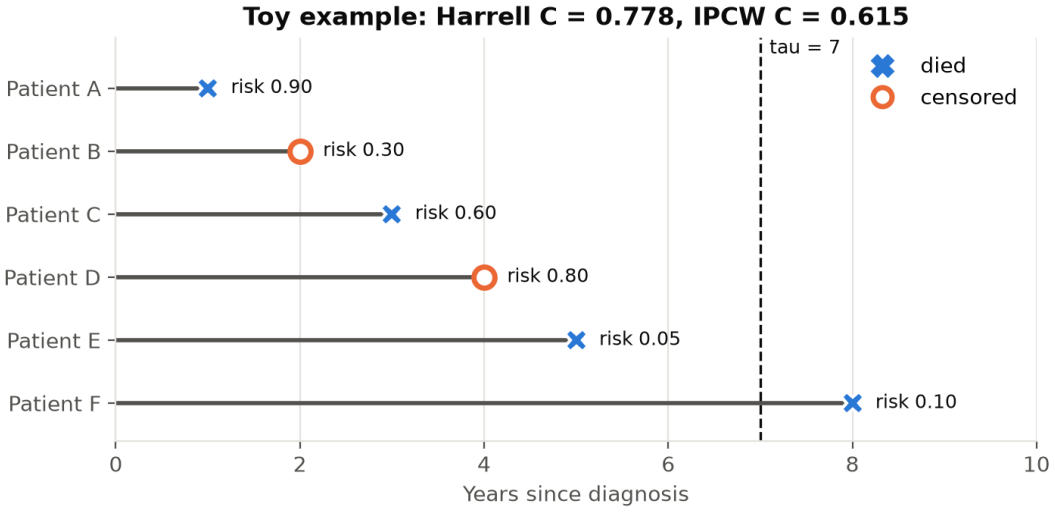


Figure 9: Toy example: crosses are deaths, circles are censored patients; each patient carries a risk score.

Comparable pairs. Only deaths can start a comparable pair, and only with patients observed for longer. Patient A (death at 1) is comparable with B, C, D, E, F; patient C (death at 3) with D, E, F; patient E (death at 5) with F; patient F (death at 8) has nobody after it. That gives

$5 + 3 + 1 = 9$ comparable pairs. Of these, two are discordant: (C, D) because $0.60 < 0.80$ and (E, F) because $0.05 < 0.10$. So

$$C_{\text{Harrell}} = \frac{7}{9} = 0.778.$$

Censoring curve. Censorings occur at $t = 2$ (B), with 5 patients at risk, and at $t = 4$ (D), with 3 at risk:

$$\hat{G}(2) = 1 - \tfrac{1}{5} = 0.8, \qquad \hat{G}(4) = 0.8 \times \left(1 - \tfrac{1}{3}\right) = 0.5333.$$

So $\hat{G}(1) = 1$, $\hat{G}(3) = 0.8$ and $\hat{G}(5) = 0.5333$, giving the weights

$$w_A = \frac{1}{1^2} = 1, \qquad w_C = \frac{1}{0.8^2} = 1.5625, \qquad w_E = \frac{1}{0.5333^2} = 3.5156.$$

Patient F died at $8 \geq \tau = 7$ and starts no pair anyway.

IPCW C-index.

$$\begin{aligned} \text{numerator} &= 5 \times 1 + 2 \times 1.5625 + 0 \times 3.5156 = 8.1250, \\ \text{denominator} &= 5 \times 1 + 3 \times 1.5625 + 1 \times 3.5156 = 13.2031, \\ C_7 &= \frac{8.1250}{13.2031} = 0.615. \end{aligned}$$

Both values match `scikit-survival` exactly. The IPCW value is lower because both wrong pairs start from later deaths (C at 3 years, E at 5 years), whose weights are larger.

5.5 Single-variable baselines

Before fitting any model, each variable can be used directly as a risk score, with its direction fixed by clinical knowledge rather than learned from the outcome (missing values set to the median). Table 9 and Figure 10 give the results on the 3,173 training patients.

Table 9: Concordance of single-variable risk scores ($\tau = 7$ years).

Risk score	Harrell's C	IPCW C
Random numbers (reference)	0.509	0.509
+ bone-marrow blasts	0.631	0.616
− platelets	0.631	0.608
+ number of mutations	0.602	0.605
− haemoglobin	0.603	0.593
+ *TP53* mutated (0/1)	0.566	0.553
− *SF3B1* mutated (0/1)	0.554	0.544
+ monosomy 7 (0/1)	0.532	0.526
− bone-marrow blasts (wrong sign)	0.369	0.384

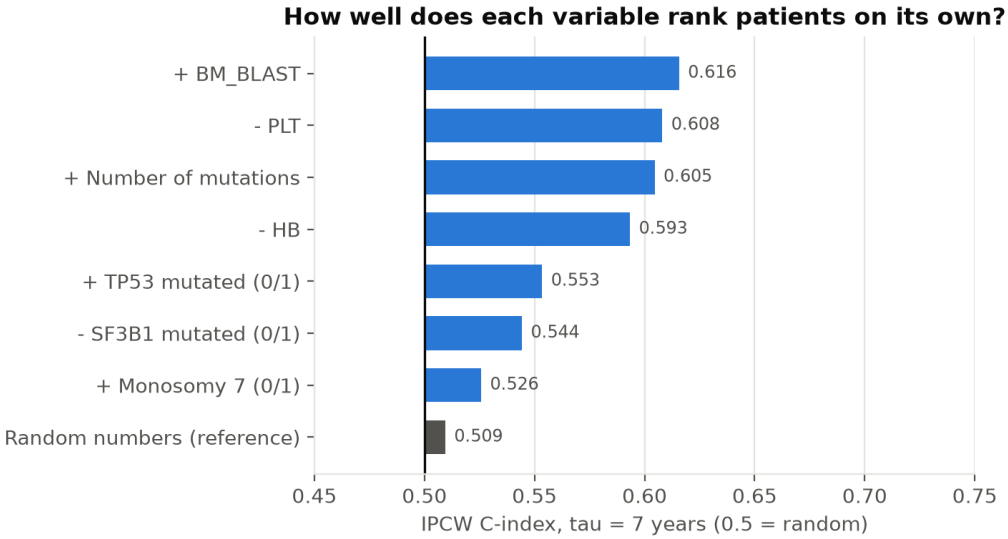


Figure 10: IPCW C-index of each single-variable risk score; 0.5 corresponds to random ranking.

Three lessons follow. First, reversing the sign of a score turns C into $1 - C$ ($0.631 \to 0.369$), so the direction of a risk score matters. Second, binary markers rank poorly on their own even when their log-rank test is overwhelming: *TP53* has $\chi^2 = 244.6$ but an IPCW C of only 0.553. Only 11.6% of patients carry the mutation, so most comparable pairs have equal scores and count one half. Statistical significance and ranking power are different things. Third, no single variable exceeds about 0.62, so combining them in a multivariable model is necessary.

6 Next steps

Stage 3 will build features: clinical transforms and missing-value handling, cytogenetic indicators parsed from the normalised karyotype text (normal karyotype, monosomy 7, deletion 5q, complex karyotype and others), and mutation features (gene indicators, number of mutations, maximum VAF). All steps that learn from data will be fitted inside cross-validation folds. Stage 4 will then fit Cox models under centre-grouped cross-validation and produce the first submission.

References

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